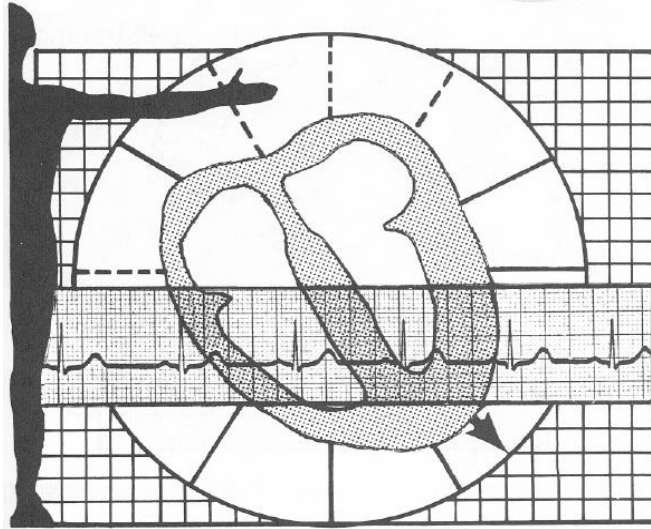
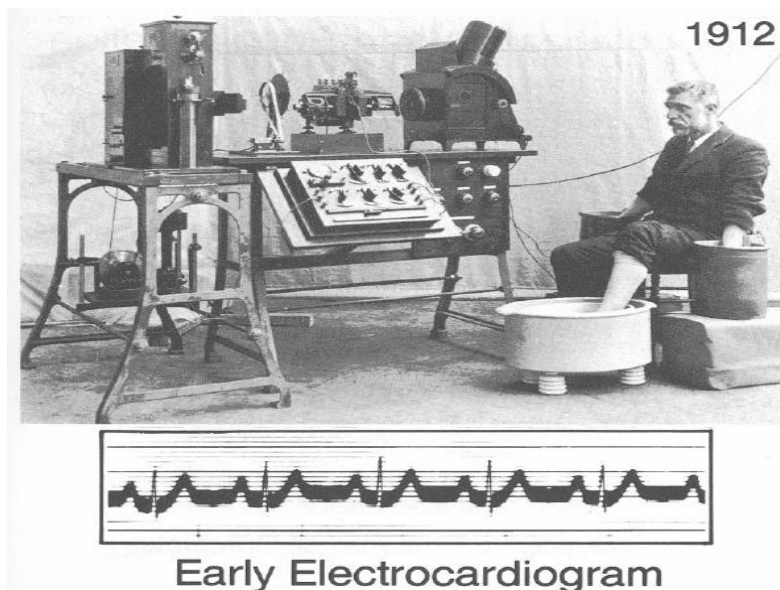


Basic Concepts and the Normal Electrocardiography (ECG)



Electrocardiogram (ECG) is a recording of **electrical activity** of the heart from the body surface. By placement of electrodes on designated areas of the body, we usually **record 12 views** of this electrical activity. Electrocardiographic monitoring provides a means of continuous observation of the electrical activity of the heart. Continuous monitoring can be used to determine conduction of impulses & irritability of tissues as well as to detect alterations in rate & rhythm. Changes in the ECG also indicate the effect of drugs and electrolytes and areas of damage to the myocardium. All **four chambers**, Lt & Rt atria & Lt & Rt ventricles, are represented on this recording.

In early days of electrocardiography, electrodes were attached to container of conducting solution (**saline**), not directly to the patient. **Willem Einthoven** is known as father of electrocardiograph.

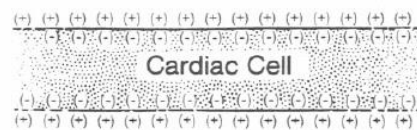


Early Electrocardiogram

Function of the heart is pumping blood for body's needs. **Mechanical act of pumping blood** is preceded by, & responsive to, **electrical stimulus**. ECG is a recording of these electrical events. In order for current to flow, there must be **positive (+) and negative (-) charges**. These are contained in & around the specialized cells of the heart.

In resting state **outside of the cell is more positive** relative to inside of cell. This is the **balanced** or **polarized** state, with no flow of electricity. When polarized cell is stimulated, polarity of cell is reversed. **Inside of cell becomes more positive** relative to outside. This process is known as **depolarization** and reflects flow of electrical current to all cells along the **pathways of conduction**. The cell then returns to its **original resting state** by process called **repolarization**.

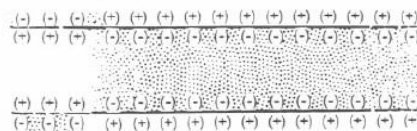
Depolarization and Repolarization



1. Resting or Polarized State



2. Depolarization, Almost Complete



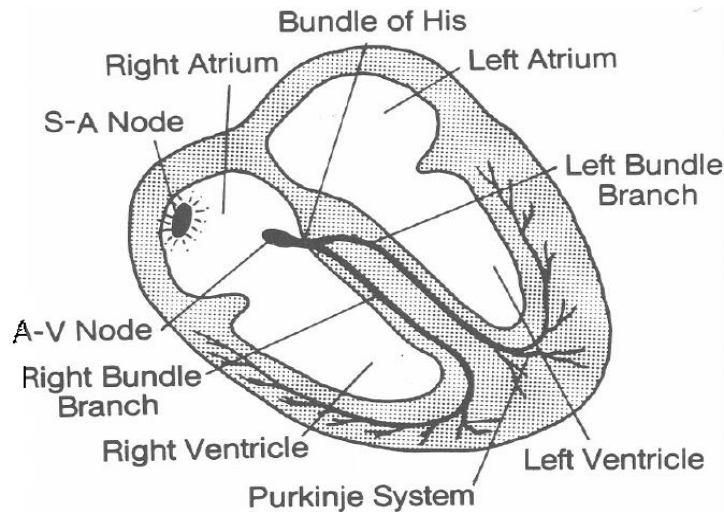
3. Repolarization, Almost Complete

Physiologic Properties of Myocardial cells

- ◆ **Automaticity** - Ability to initiate an impulse.
- ◆ **Excitability** - Ability to respond to an impulse
- ◆ **Conductivity** - Ability to transmit an impulse.
- ◆ **Contractility** - Ability to respond with pumping action.

Electrical Conduction of the Heart

Process of conduction of impulse within heart is known as its **electrical conduction system**. **Sinoatrial (S-A) node** is **normally the site of origin of the electrical impulse**, leading to depolarization of atria. Impulse then spreads through the **atrioventricular (A-V) node** and **bundle of His** to the **left (LBB)** and **(RBB) bundle branches** & then to **ventricles** through **Purkinje fiber network**, leading to ventricular depolarization.

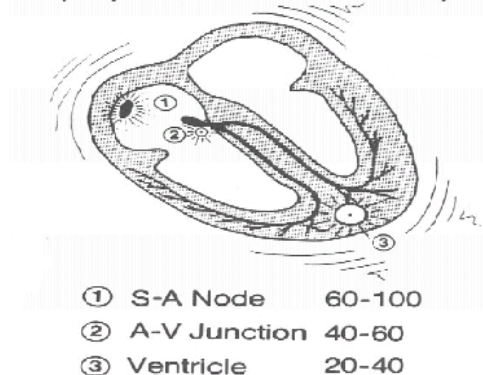


Although the **primary & dominant pacemaker** of heart is **S-A node** under various circumstances and stimuli, **another pacemaker** may become dominant there may be **two or more pacemakers** propagating impulses at the same time. Each pacemaker has its own **inherent rate**.

In general, pacemaker with **fastest inherent rate** becomes dominant cardiac pacemaker. **S-A node** emits **60-100 impulses/min**. **A-V junction** emits **40-60 impulses/min** and still **lowers pacemakers**, **20-40 impulses/min**. Thus, S-A node is usually the fastest & dominant cardiac pacemaker.

If **secondary pacemaker**, such as **junctional** or **ventricular pacemaker**, **speeds up**, it may become dominant pacemaker. Also, if S-A node slows down or fails, secondary pacemaker may become dominant.

Rate of Impulse Formation (Impulses Per Minute)



Autonomic Nervous System: Sympathetic & Parasympathetic Nervous

Sympathetic System

- ◆ Supplying both atria & Ventricles

Mediator - Norepinephrine

Increases:

- Rate of SA node
- Rate of AV conduction
- Excitability
- Force of contraction.

Parasympathetic system

- ◆ Supplying Atria Primarily

Mediator - Acetylcholine

Decreases:

- Rate of S-A node
- Rate of A-V contraction
- Excitability

✚ If one system is blocked, the effects of other are seen. For e.g., if SNS is blocked, effects of PNS would be elicited.

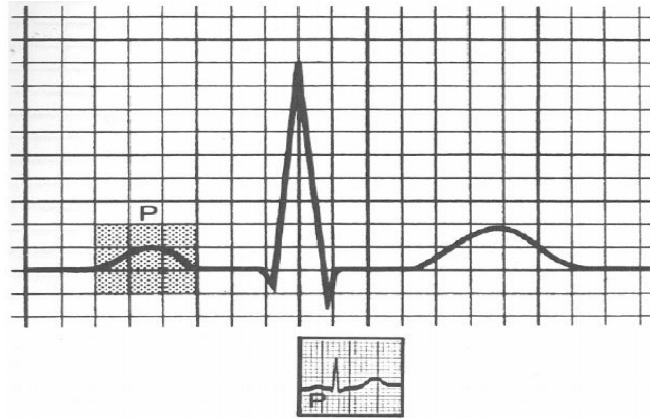
Normal ECG: All beats appear as a normal pattern, equally spaced, and have three major units: **P wave**, **QRS complex**, and **T wave**.

Normal single heart beat: Each beat manifests as five major waves: **P, Q, R, S, & T**. The Q, R, & S all represent the same part of the heart (ventricles). They are usually referred as a unit: the QRS complex.

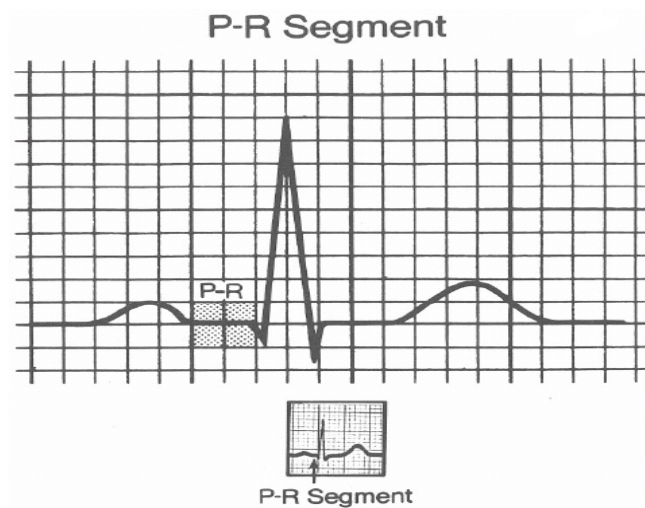
Atrial depolarization

Wave of depolarization that begins in S-A node spreads to both atria, first to the Rt atrium, then to the Lt atrium. Depolarization of both atria is represented by the P wave on ECG.

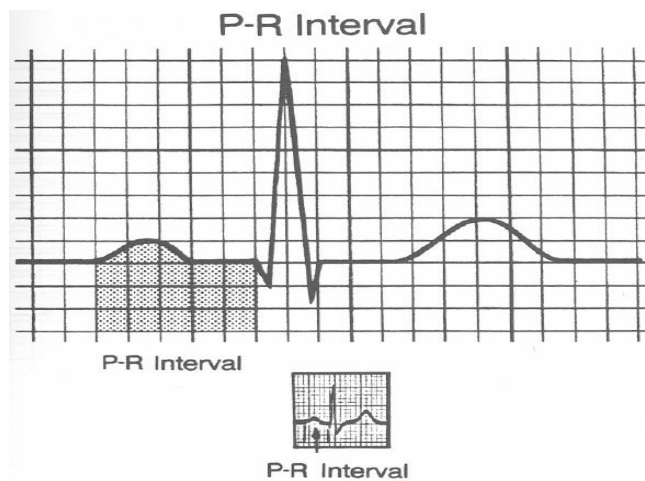
P wave is normally the first upward deflection of each cardiac cycle. Enlargement of the P wave might be occur in such conditions as *mitral stenosis* of chronic obstructive pulmonary diseases, which would cause *atrial hypertrophy*. P wave is considered enlarged if it is more than 3 small boxes (3 mm) high and/or 3 small boxes (0.12 sec.) wide.



During **P-R segment**, following atrial depolarization, electrical impulse spreads to A-V node, bundle of His, and bundle branched.

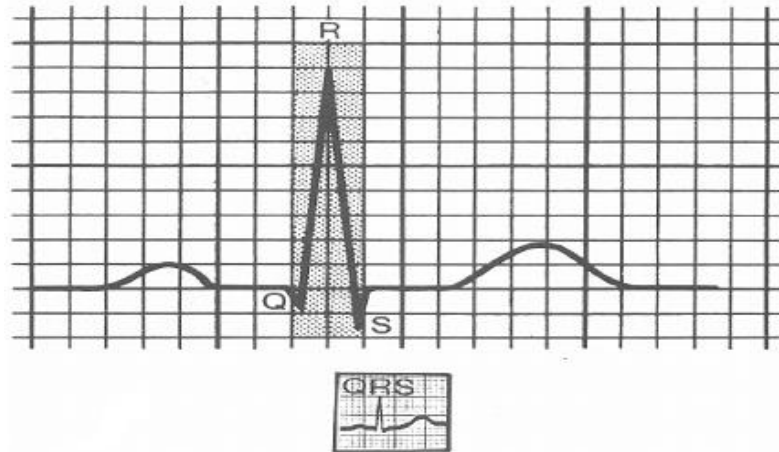


P-R interval includes **P wave** & **P-R segment**, represents time of transmission of electrical impulse from **atria** to **ventricles**. **Normal P-R interval** is 3-5 small blocks wide (0.12 - 0.2 sec.).



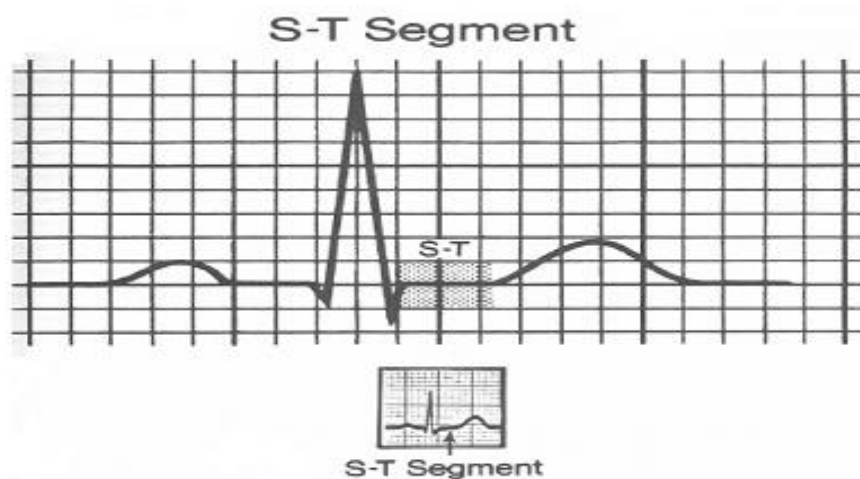
Depolarization of both ventricles is reflected in **QRS complex**. **R wave** is the **initial positive deflection** (upward above baseline of ECG) of the QRS complex. **Negative deflection** (downward, inverted, and occurring below resting baseline of ECG) before R wave is **Q wave**. Negative deflection after R wave is **S wave**, which is usually part of QRS complex. **Normal duration of the QRS** is 2 & 1/2 small boxes or less.

Ventricular Depolarization



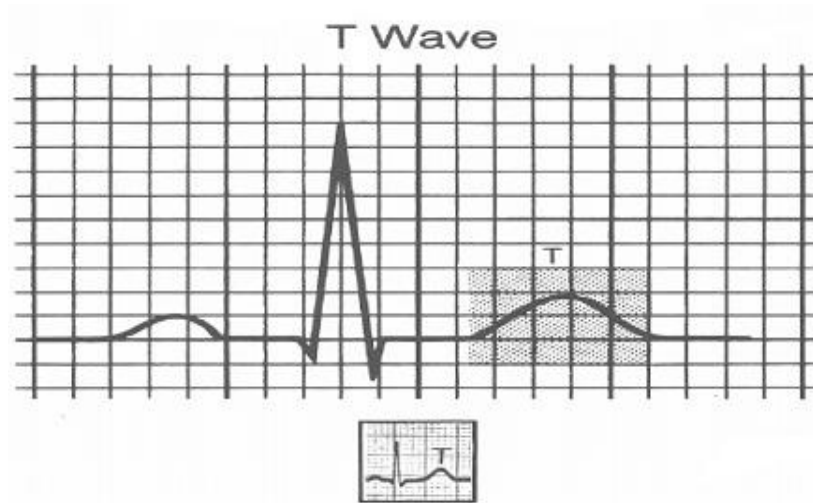
S-T segment extends from QRS complex to beginning of T wave. It represents earlier **phase of repolarization** of the ventricles. **S-T segment** is normally **isoelectric** (at same level as resting baseline). It is neither **elevated** (positive) nor **depressed** (negative). The point at which S-T segment joins QRS complex is known as **J point**. It is elevated in an acute MI or muscle injury. It is depressed when the heart is not getting a sufficient supply of oxygen, e.g. coronary insufficiency.

Ventricular Repolarization



Later repolarization of both ventricles inscribes the **T wave** on electrocardiogram. S-T segment & T wave are **sensitive indicators** of status of ventricular myocardium. The normal is no more than 10 small blocks (10 mm) high in the precordial (chest) leads and 5 small blocks (5 mm) high in the remaining leads. T wave is flat or inverted in response to ischemia, position change, or certain drugs. It might be elevated when the serum potassium is elevated.

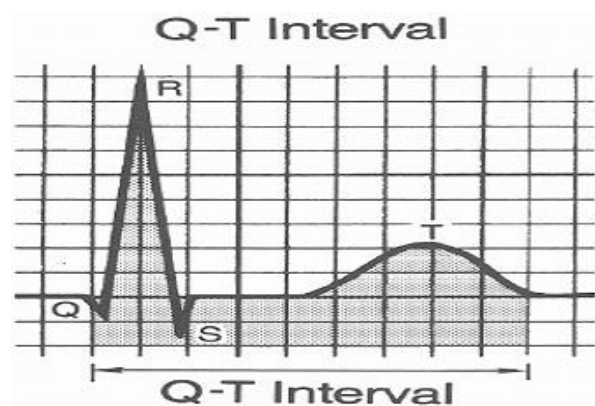
Ventricular Repolarization



Q-T interval includes QRS complex, S-T segment, & T wave. Q-T interval extends from the beginning of Q wave to end of T wave. It represents **ventricular depolarization & repolarization**.

U wave is a small upward deflection following the T wave. It is seldom present, but occurs when the serum potassium level is low.

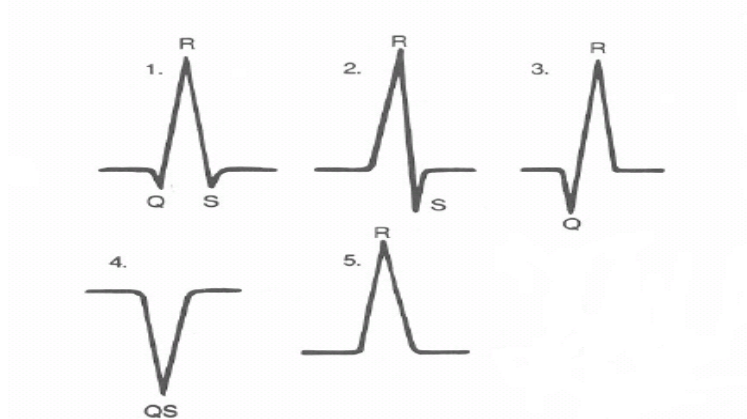
Ventricular Depolarization and Repolarization



Types of Ventricular Complexes

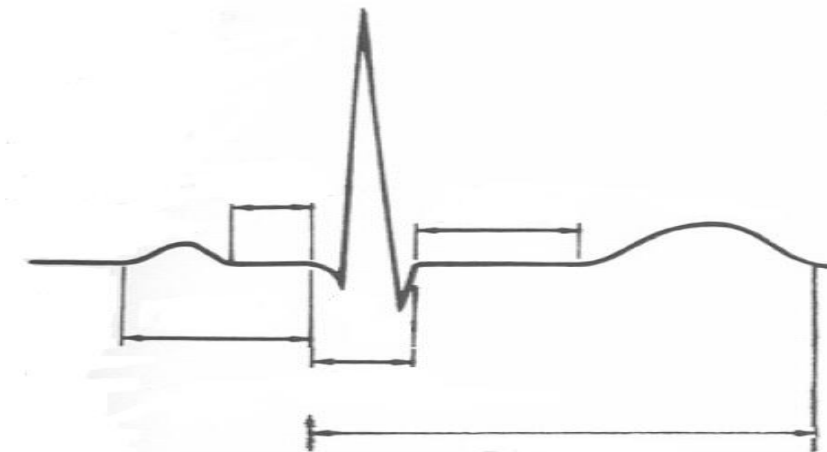
1. **QRS**; Q wave - negative deflection before the R wave
R wave - positive deflection
S wave - negative deflection after the R wave
2. **RS**; no Q wave present
3. **QR**; no S wave present
4. **QS**; totally negative complex, no R wave present
5. **R**; no Q wave or S waves present

Types of Ventricular Complexes (QRS)*

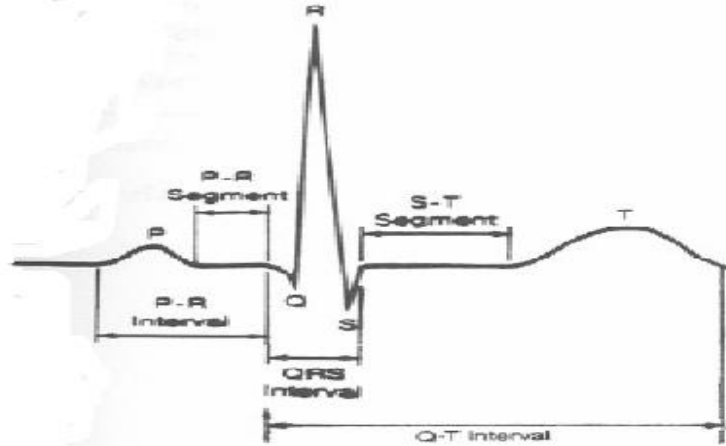


Practice

Identify the principal electrocardiographic waves (P, QRS, and T)?



Answer



Thorough familiarity with the nomenclature of the components of the cardiac cycle is vital in electrocardiography. Prior to studying the placement of the leads, heart rate determination, and measurements, review all materials studied so far.

The ECG leads

To record a routine ECG, 12 leads are used:

On the **limbs**: I, II, III, AVR, AVL, and AVF

On the **chest**: V₁, V₂, V₃, V₄, V₅, V₆

Each lead records the same electrical impulse on the ECG but from a different position in relation to the heart. Each can be read separately, or they can be read in combination.

The limb leads show the current flow from one area of the body toward another. Lead I shows electrical activity from the Rt arm to the Lt arm, lead II from Rt arm to Lt leg, and lead III from Lt arm to Lt leg.

The abbreviation AVR stands for augmented vector right, an added point of reference for diagnosis. AVL and AVF represent the left-side and left-foot positions respectively.

In **diagnosing arrhythmias**, lead II and the right-chest leads are commonly used because they most consistently show an easily visible P wave. Many times, the P wave provides the key to success in determining the identity of an arrhythmia.

The most important leads to remember in relation to the anatomy of the heart are:

V₁, AVR.....Right side of the heart.

V₂, V₃, V₄.....Transition between right and left.

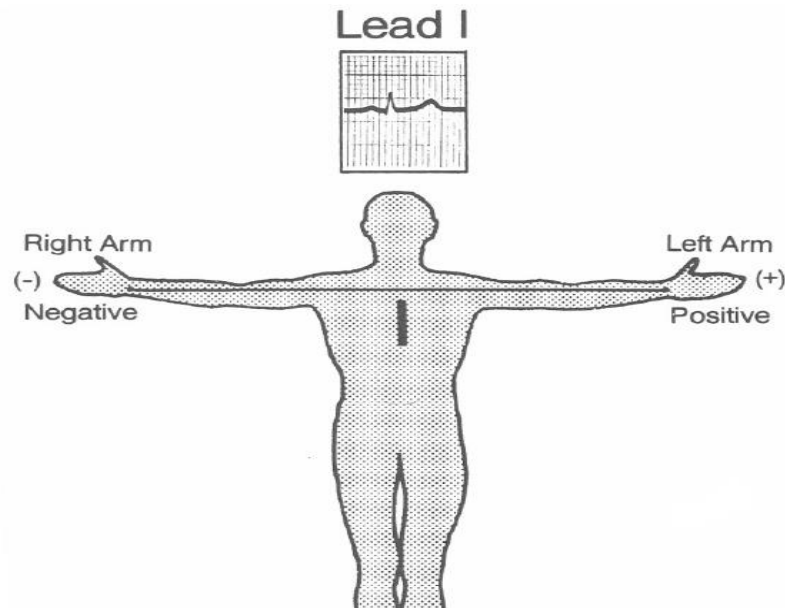
V₅, V₆, I, AVL..... Left side of the heart.

II, III, AVF..... Inferior aspect of the heart.

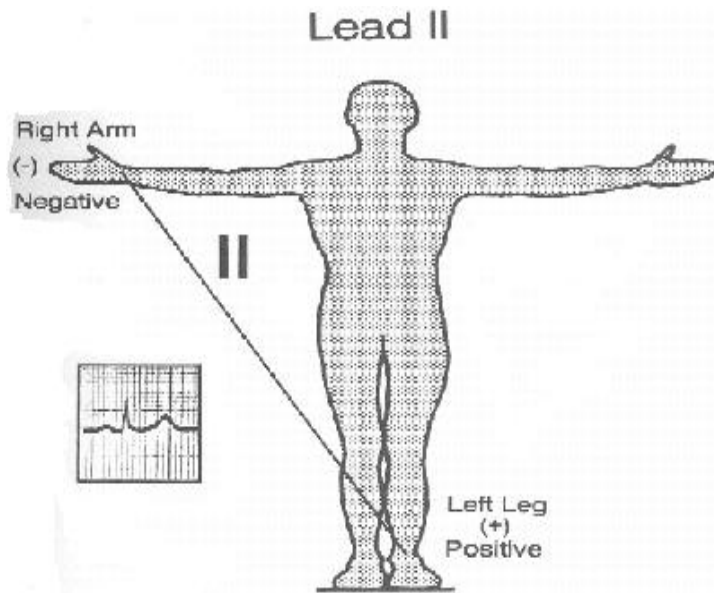
The **area of pathology** shown in the ECG can be localized by analyzing tracing from different leads. For e.g. if an infarct shows up in leads II, III, and AVF only, it is located in the inferior aspect of the heart.

Limb Leads

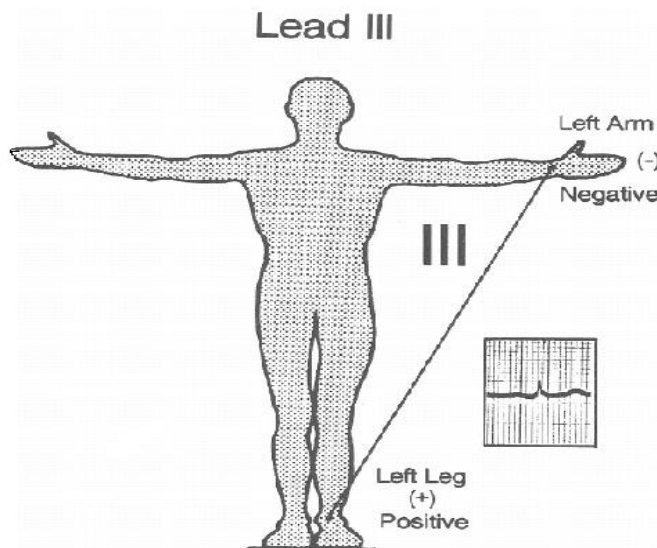
Einthoven found that by connecting Rt arm to negative electrode & Lt arm to positive electrode and inducing a current to flow, certain deflections were recorded, known as lead I. For recording, Einthoven used string galvanometer, a movable writing element within a magnetic field.



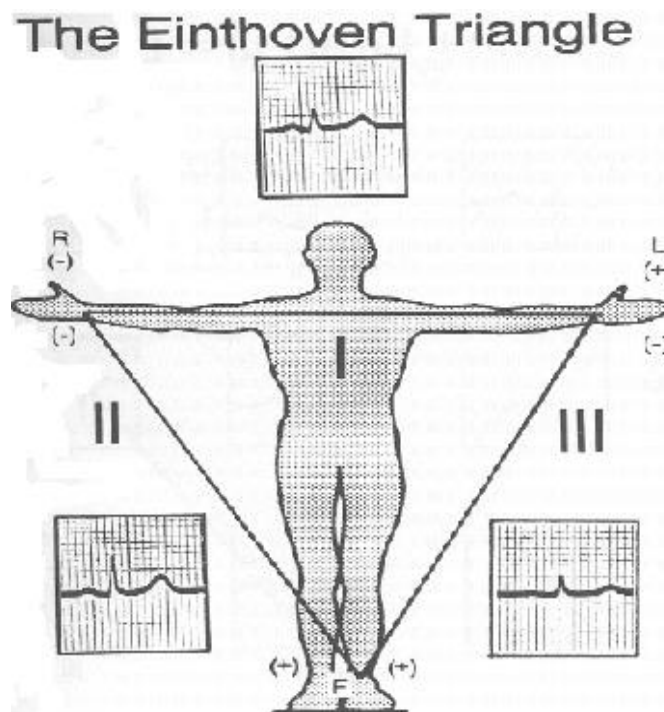
By connecting Rt arm & Lt leg, Rt arm to negative electrode & Lt leg to positive electrode, Einthoven recorded what become known as **lead II**.



By connecting Lt arm & Lt leg, Lt arm to negative electrode & Lt leg to positive electrode, **lead III** is formed.

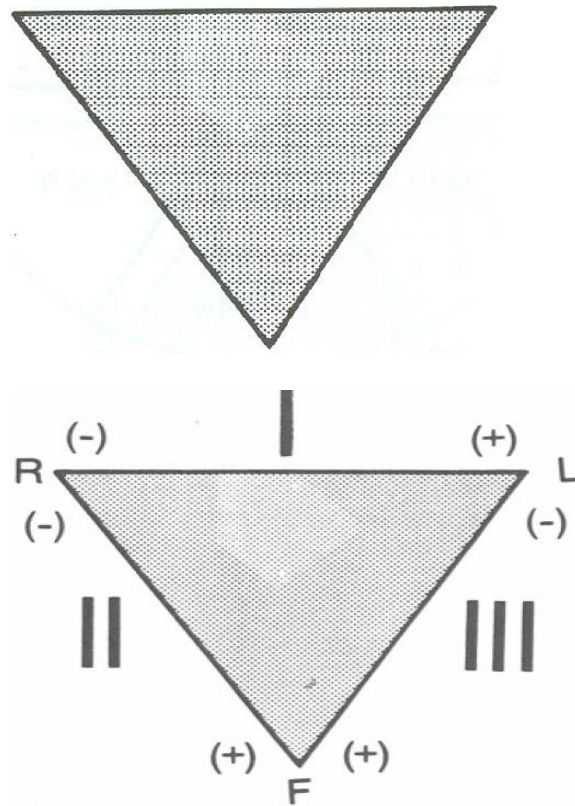


These three leads, arranged as above, are known as the **Einthoven triangle**. Rt arm is always **negative** & Lt leg is always **positive**. Einthoven triangle is composed of leads I, II & III.



Practice

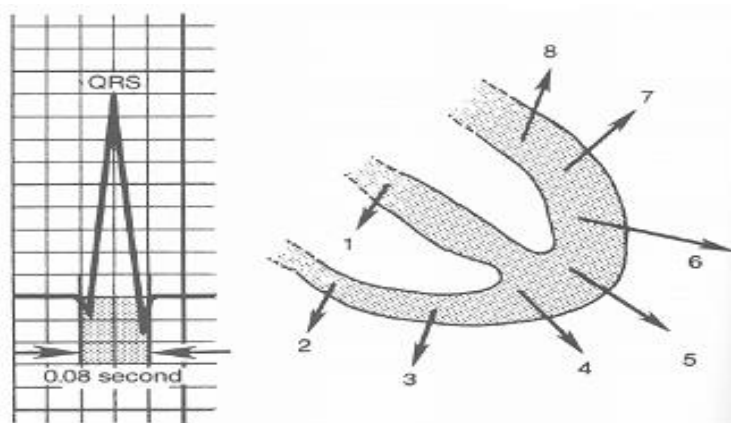
Label the Einthoven triangle, including the polarity (positive or negative) of the leads?



Answer:

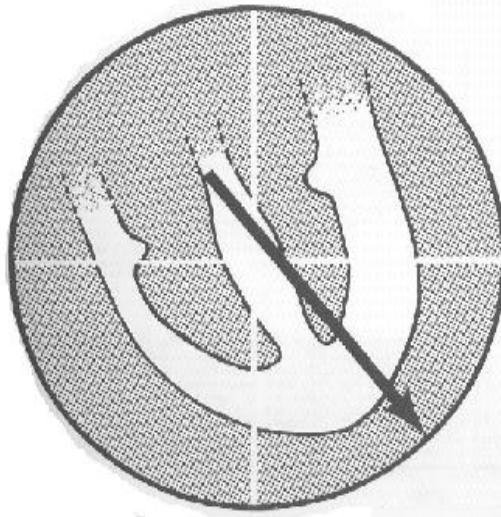
The Cardiac Vector

The term vector refers to force. In electrocardiography it refers to electrical force. A vector is represented by an arrow where both size & direction are easily demonstrated within a period of 0.08 sec. both ventricles are depolarized. During this single **QRS** interval, sequential instantaneous vectors are generated. The greater the muscles mass, the bigger the arrow will be.



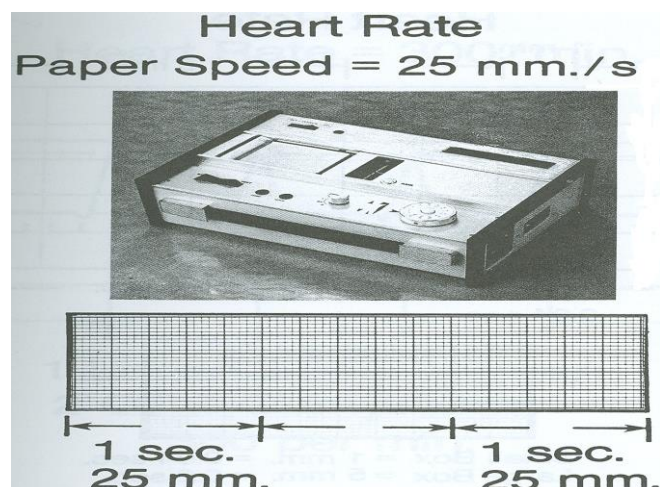
The **mean QRS vector** is the resultant, or mean vector, of all the instantaneous vectors within a single QRS interval. We often refer to mean **QRS vector**, mean **T vector**, & mean **P vector** in the study of electrocardiography. Because of common usage, the terms **mean electrical axis**, **electrical axis**, or simply "**axis**" of the QRS, T & P are used.

Mean QRS Vector
Mean Electrical Axis of QRS
"Axis of QRS"

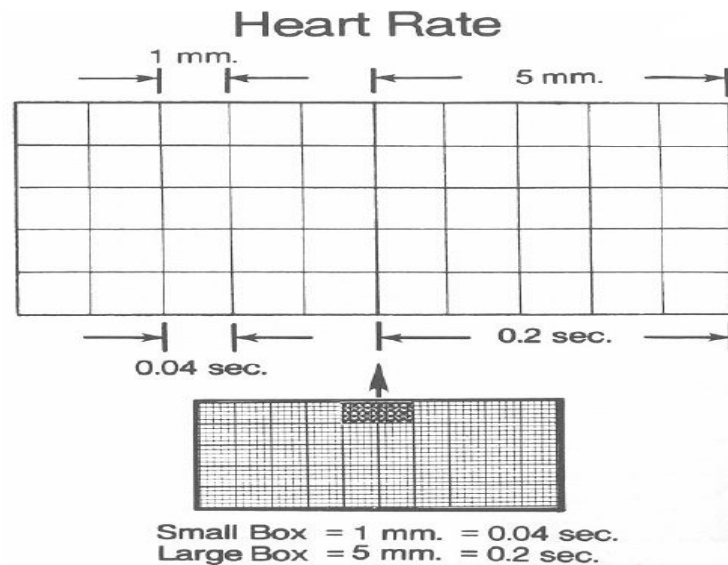


Determination of the Heart Rate from the Electrocardiography

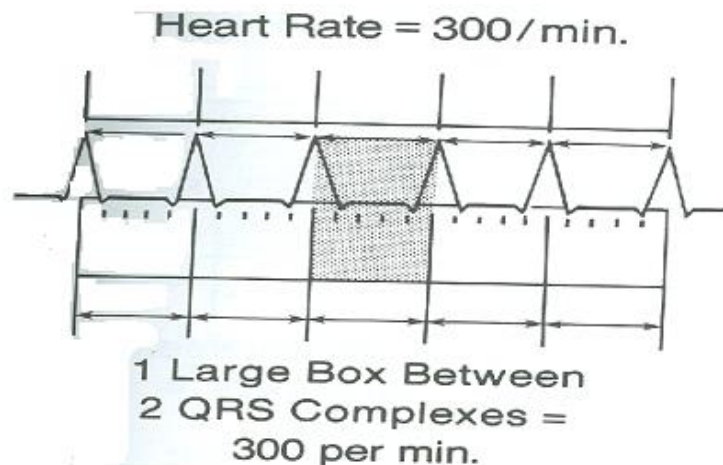
Determination of HR from ECG depends on the spread of the paper when recording the ECG. The standard speed of the paper is **25 mm. (five large boxes) per sec.** whenever the paper speed is changed, which is possible on most machines, it should be indicated on the electrocardiogram.



The ECG is recorded on **lined paper** that consists of large & small boxes. *Each large box measures 5mm. & each small box measures 1mm.* at the paper speed of 25 mm. per sec., *each large box represents 0.2 sec. & each small box 0.04 sec.*



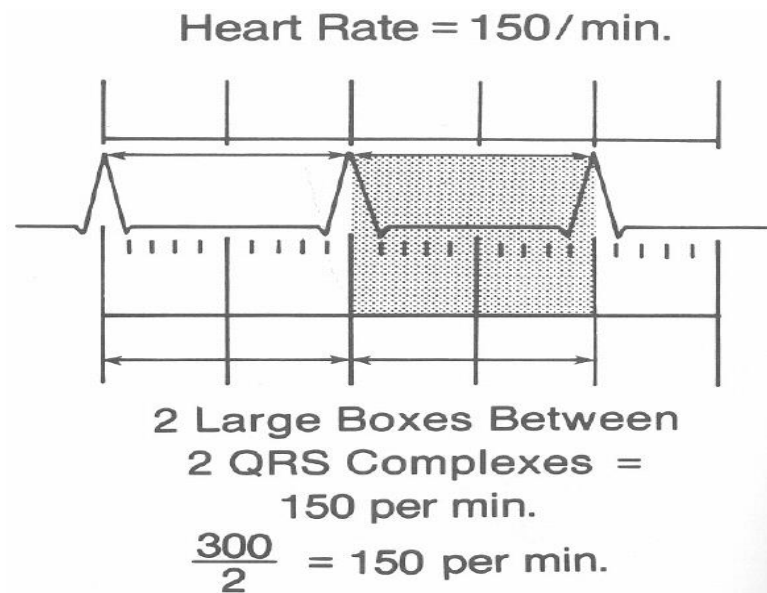
- ♦ At the usual paper speed of 25 mm/sec (five large boxes), when HR is **300 per min.** the interval is **one large box (5 mm, 0.2 sec.) between two complexes.** All that is necessary to determine HR when the **rhythm is regular** is to count the number of large boxes between two QRS complexes & divide into 300.



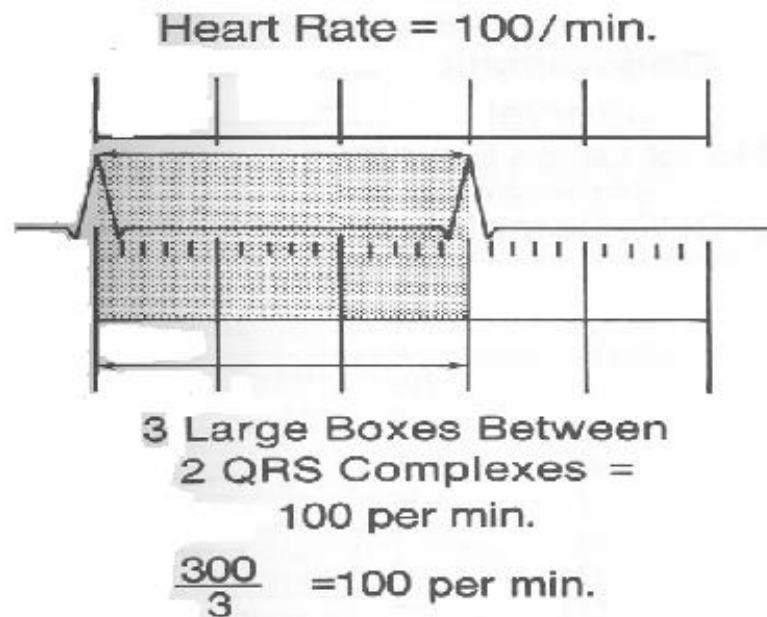
Note: - To determine HR when the rhythm is regular, two QRS complexes are needed.
 - At standard ECG paper speed, one large box b/n 2 QRS complexes equals a rate of 300 per min.

- ♦ When HR is 150/min, the interval is **2 large boxes (0.4 sec.) between two QRS complexes (300/2).**

This determination is facilitated if you start the count with a QRS complex or P waves, whichever you are measuring, that falls on a heavy line.



- ♦ When HR is 100/min, the interval is **3 large boxes (0.6 sec.) between two QRS complexes (300/3).**



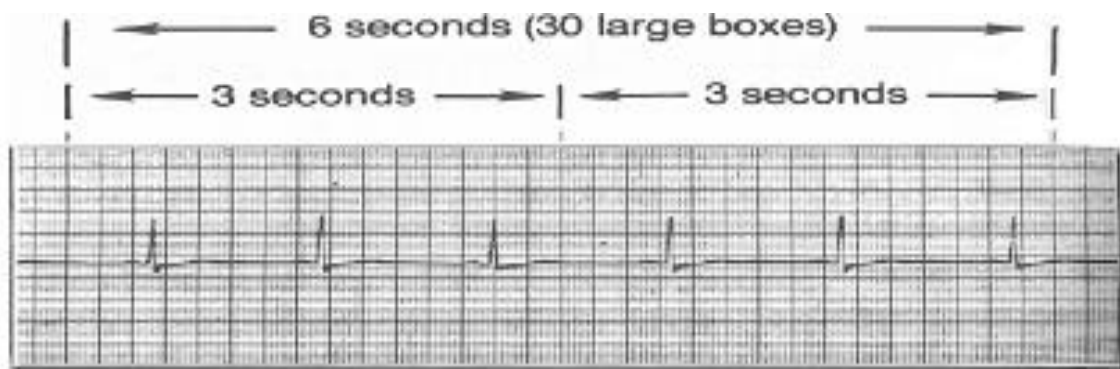
Practice

Determination of Heart rate

<u>Interval</u>		<u>Rate/min</u>
(No. of large boxes		
Between 2 QRS complexes)		
1	=	_____
2	=	_____
3	=	_____
4	=	_____
5	=	_____
6	=	_____
7	=	_____
8	=	_____

This relationship permits the determination of HR using only two QRS complexes when the **rhythm is regular**. Simply count the number of large boxes & any fraction, & divide into 300 for the HR. when the rhythm is irregular, you have to count numerous QRS complexes to arrive at the proper average.

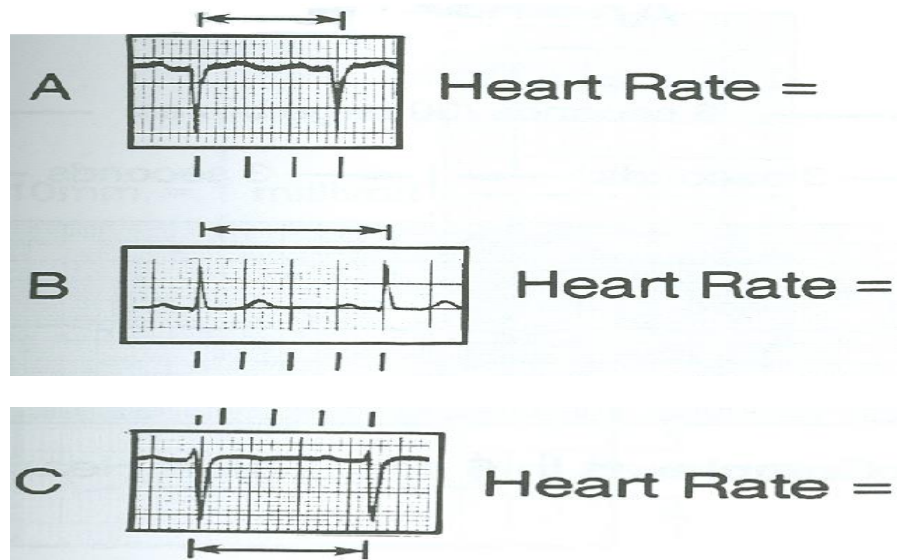
Heart Rate Determination Another Method



QRS Complexes in 6 sec. $\times 10$ = Heart Rate

Another method of determining HR is to utilize the 3-sec. markers (15 large boxes = 3 sec.) at the upper border of the electrocardiographic paper. **Count the number of QRS complexes in a 6 sec. period** & multiply by 10 for the rate per min. Within the 6sec. period above are six QRS complexes. Therefore the Hr is $6 \times 10 = 60$ per min. This method is especially useful when the rhythm is **not regular**.

Practice:

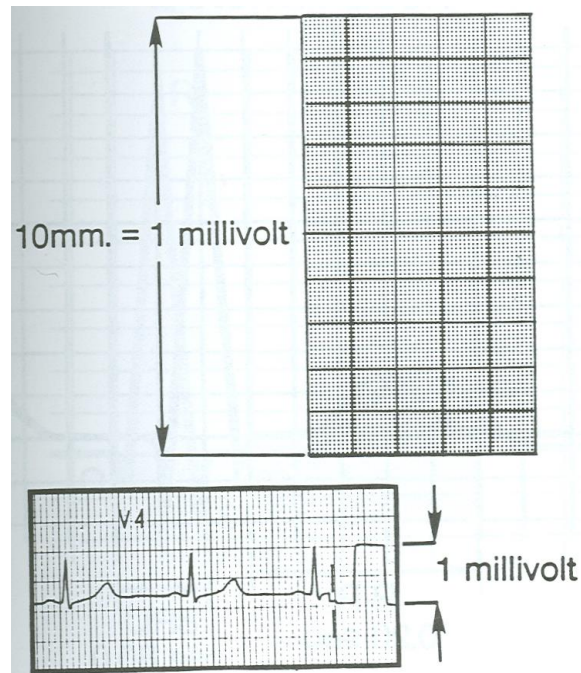


Answer:

- A. Three large boxes b/n 2 QRS complexes: HR = 100 per min. ($300 \div 3 = 100$).
- B. Four large boxes b/n 2 QRS complexes: HR = 75 per min. ($300 \div 4 = 75$).
- C. Three large and a half large boxes b/n 2 QRS complexes: HR = 86 per min. ($300 \div 3.5 = 86$).

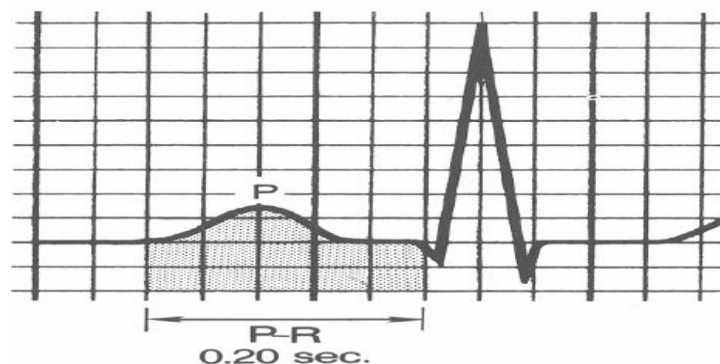
Electrographic Standardization

The usual **standardization** on the electrocardiogram of 1 millivolt results in a deflection of **two large boxes (10 mm)**. Standardization is essential to properly evaluate the size of deflection even if the ECG has been enlarged or reduced in size.

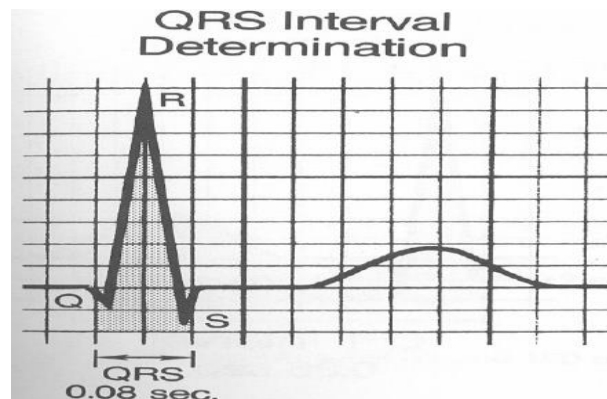


The **P-R interval** (from beginning of P wave to beginning of QRS complexes) is 0.20 sec. (five small boxes or one large box). **Normal P-R interval** is from 0.12 to 0.20 sec. This is important measurement, since an **abnormal prolongation** represents a delay in the transmission of the electrical impulse from the atria to the ventricles. A **shorter than normal P-R interval** (0.1 sec. or less) has been associated with various rhythm disturbances.

P-R Interval Determination



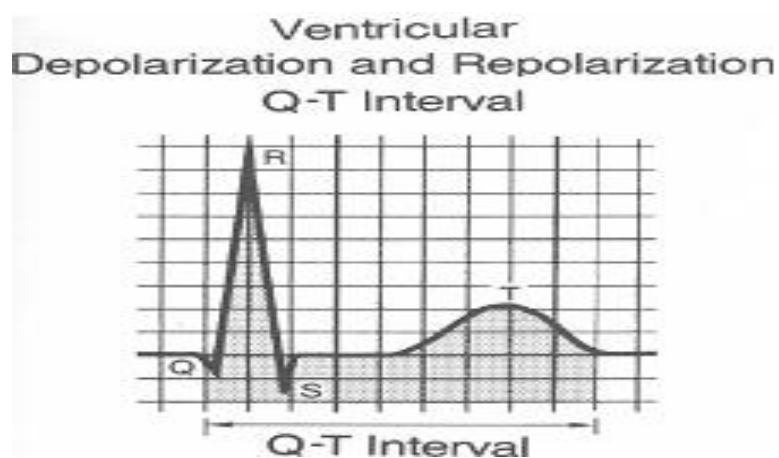
The **QRS interval** is 0.08 sec. (2 small boxes). **Normal interval** for ventricular depolarization is up to 0.1 sec. **Abnormal prolongation** represents an intraventricular conduction disturbance. **QRS interval prolongation** is seen in right & left bundle branch block.



The **ventricular repolarization process is longer at the lower rates**, which is reflected in the **longer Q-T interval** at the **lower rates**.

<u>Heart Rate (Beats per min.)</u>	<u>Q-T</u>
40	0.46
60	0.39
80	0.35
120	0.31
140	0.26
160	0.25

Note: The Q-T interval is rate dependent is decreased at the higher rates



Normal Sinus Rhythm

In order to describe the **rhythm of heart as sinus rhythm** (impulse originating in sinoatrial node, which is the normal pacemaker of the heart without qualifications, the following requirements must be met.

1. Normal Mean P Vector (to the left and inferior - upright P waves in leads I and aVF or I & II.
2. Each P wave must be followed by a QRS complex and each QRS complex must be preceded by a P wave.
3. The normal P-R interval (from the beginning of the P wave to the beginning of the QRS complex) is rarely greater than 0.2 sec. (one large box) in duration. The normal range is 0.12 to 0.2 sec. and is constant from beat to beat.
4. The rate is constant between 60 and 100 beats per minute.

Electrocardiographic Interpretation

1. Rhythm and Rate
 - P-R Interval
 - P Wave Abnormalities
 - Abnormalities of Rhythm
2. QRS Complex
 - Duration
 - Mean QRS Vector, Mean Electrical Axis or "Axis"
 - Abnormalities
3. S-T Segment and T Wave (Ventricular Repolarization).
 - QRS-T Angle
 - Abnormalities
4. Q-T Interval

Impression and Comment